CO-1 AT-3

1. A Smart home automation System is being designed to manage lighting, temperture, and Security based on user behavior and environmental conditions. The System must respond to real-time inputs such as motion detection temperature changes, and Time of day while also learning user Perference over time. In this context explain the different types of intelligent agents

A) Intelligent Agents in Smart Home.

* Simple Reflex Agent:
Acts only on the current situtation. Example: if motion is detected, turn on the light

* Model Based Agent:
Maintains information about the environment.
Example: remembers room temperature and Previous conditions.

* Goal Based Agents:
Takes action to achieve a specific goal. Example: maintains a comfortable temperature

* Utility-Based Agent:
Chooses the best action based on usefulness.
Example: balances confort, energy Saving and Security

Best choice: A hybrid agent combining model-based goal-based, and utility-based approaches is most effective. It can learn user Preference, Save energy, maintain comfort and improve security

2. A robot is Placed in an unknown maze and must find a Path to the exit without Prior knowledge of the layout. Explain how uninformed Search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help Solve this Problem.

A) UCS and IDS in a Maze

Uniform Cost Search (UCS):

- Expands the Path with the lowest cost.
- Finds the least-cost Path when costs are different

Advantage: Complete and optimal.

disadvantage: Can use a lot of time and memory.

Iterative Deeping Search (IDS):
Searches depth 0, then depth 1, depth 2 and so on
Advantage: Uses less memory and is complete
disadvantage: Repeats searches, so it can take more time.

Best combination: A model-based / goal based agent with UCS is suitable when Path cost matters. The model-based agent can remember explored maze information, while UCS finds the lowest-cost Path

code of conduct::
I M. vishvanandh 1965260005 certify that this submission is my original work and that I have adhered to the guidelines specified for this assignment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student:
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